

# DISTRIBUTED ARCHITECTURES

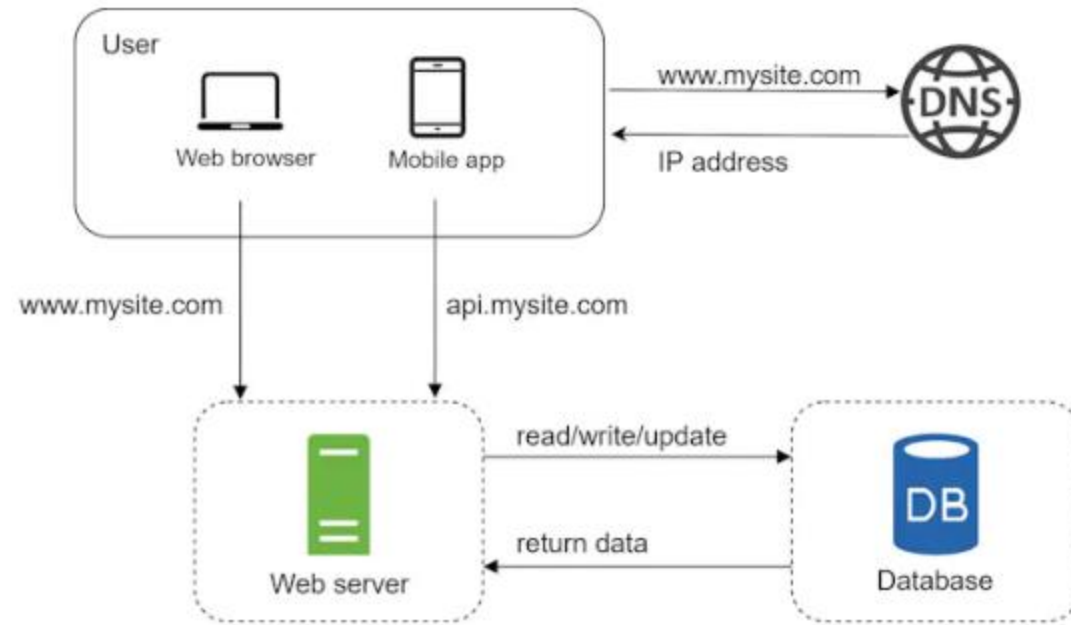
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By Jorge Galvis

# SCALE FROM ZERO TO MILLIONS

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One thing is to serve a backend for a couple of users and another one is to serve it for millions of users.

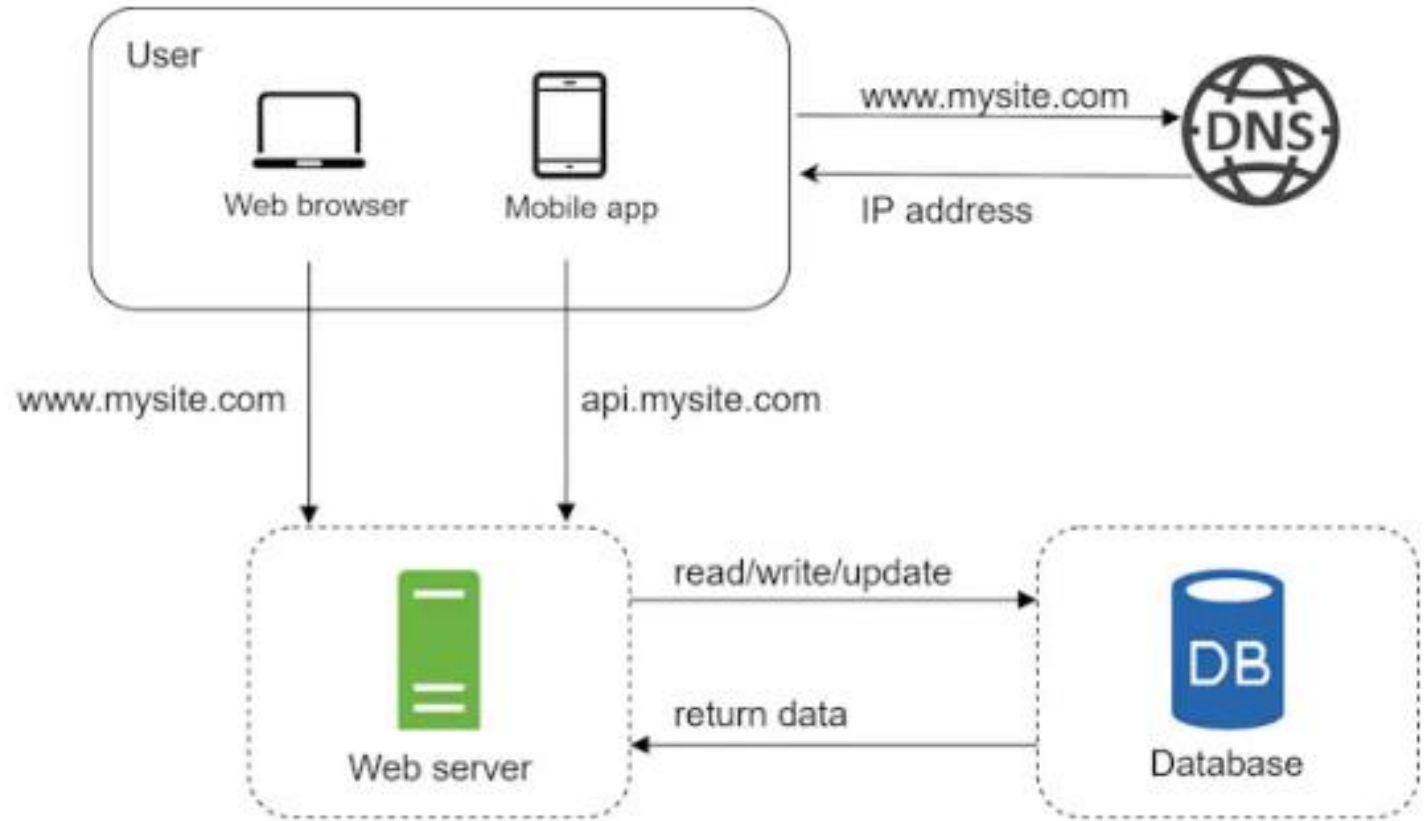


# SCALE FROM ZERO TO MILLIONS

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Databases

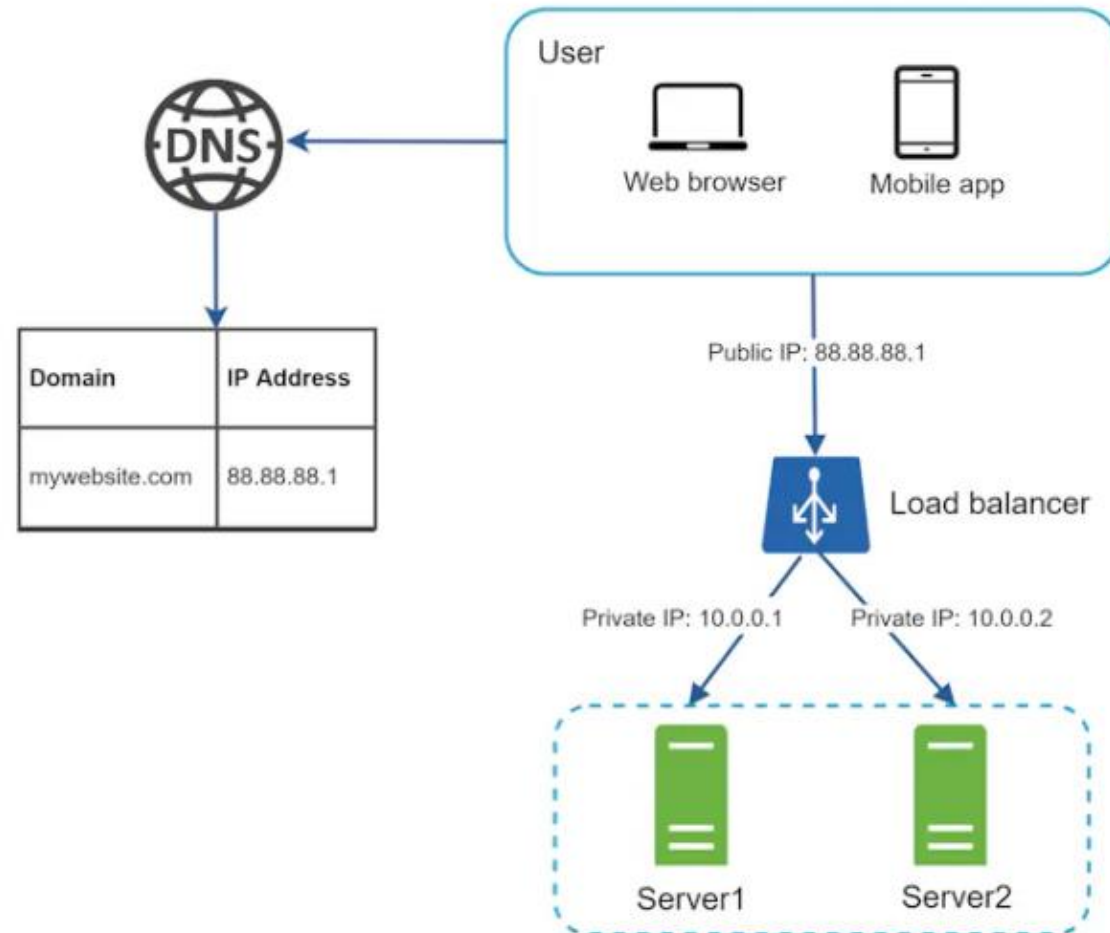
Vertical vs horizontal  
scaling



# SCALE FROM ZERO TO MILLIONS

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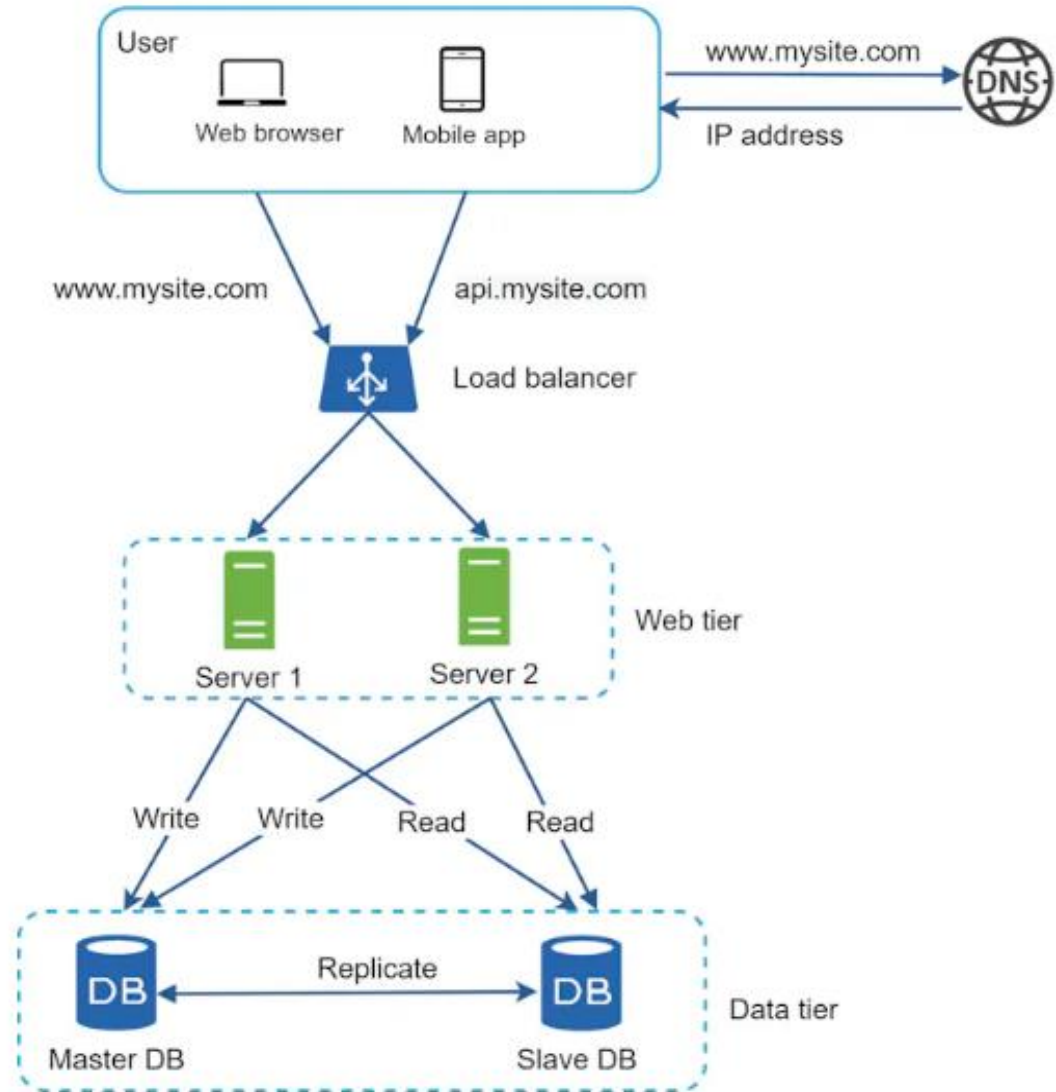
## Load Balancer



# SCALE FROM ZERO TO MILLIONS

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## Database replication

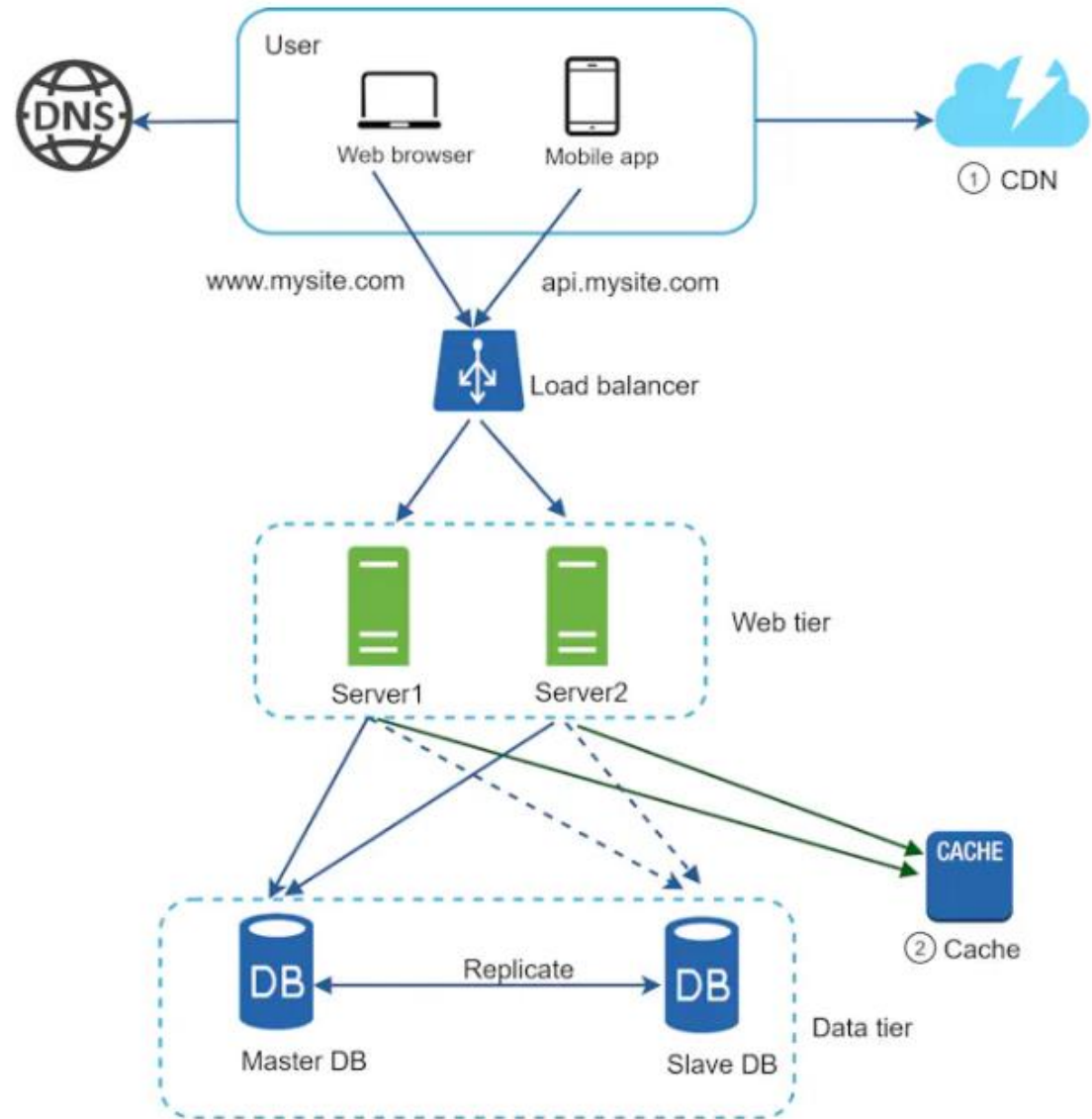


# SCALE FROM ZERO TO MILLIONS

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Cache

CDN

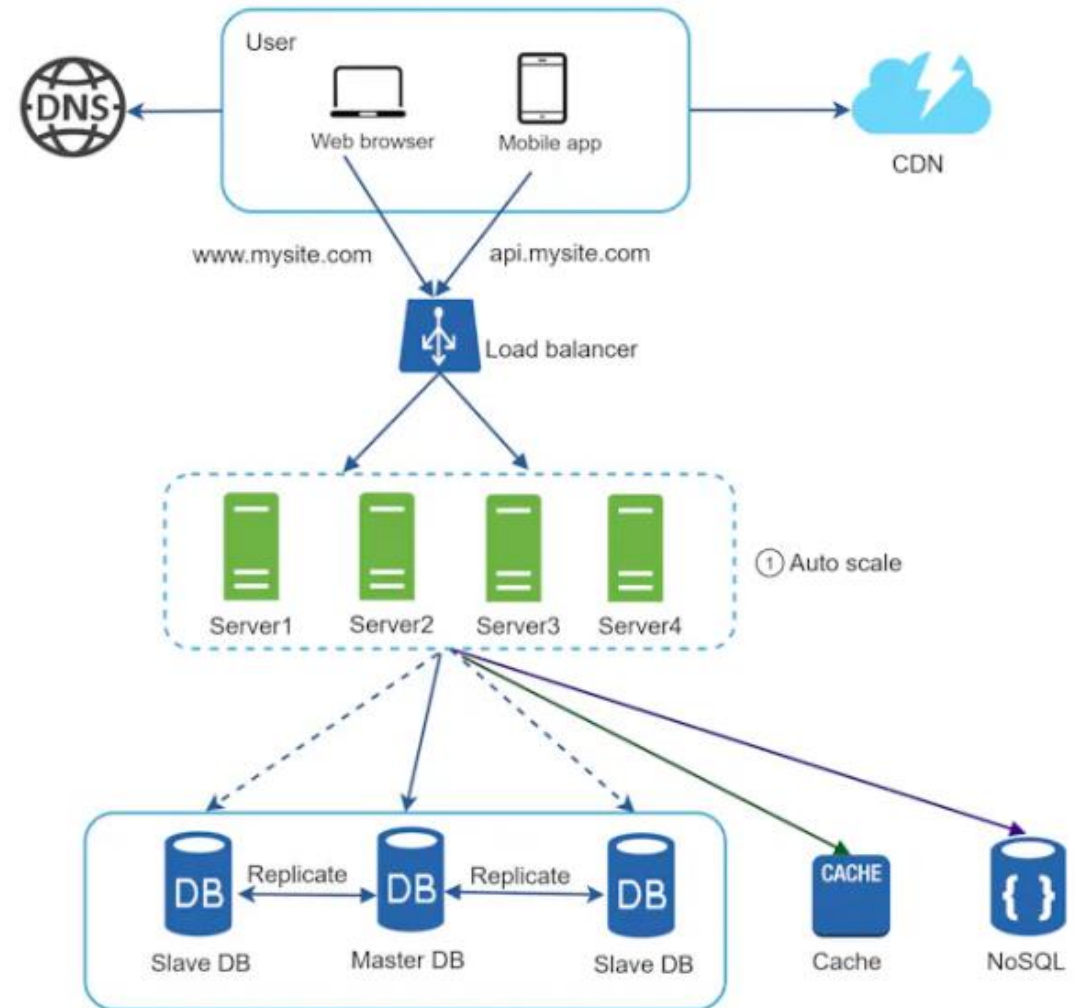


Taken from <https://bytebytego.com/courses/system-design-interview/scale-from-zero-to-millions-of-users>

# SCALE FROM ZERO TO MILLIONS

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Auto-scale

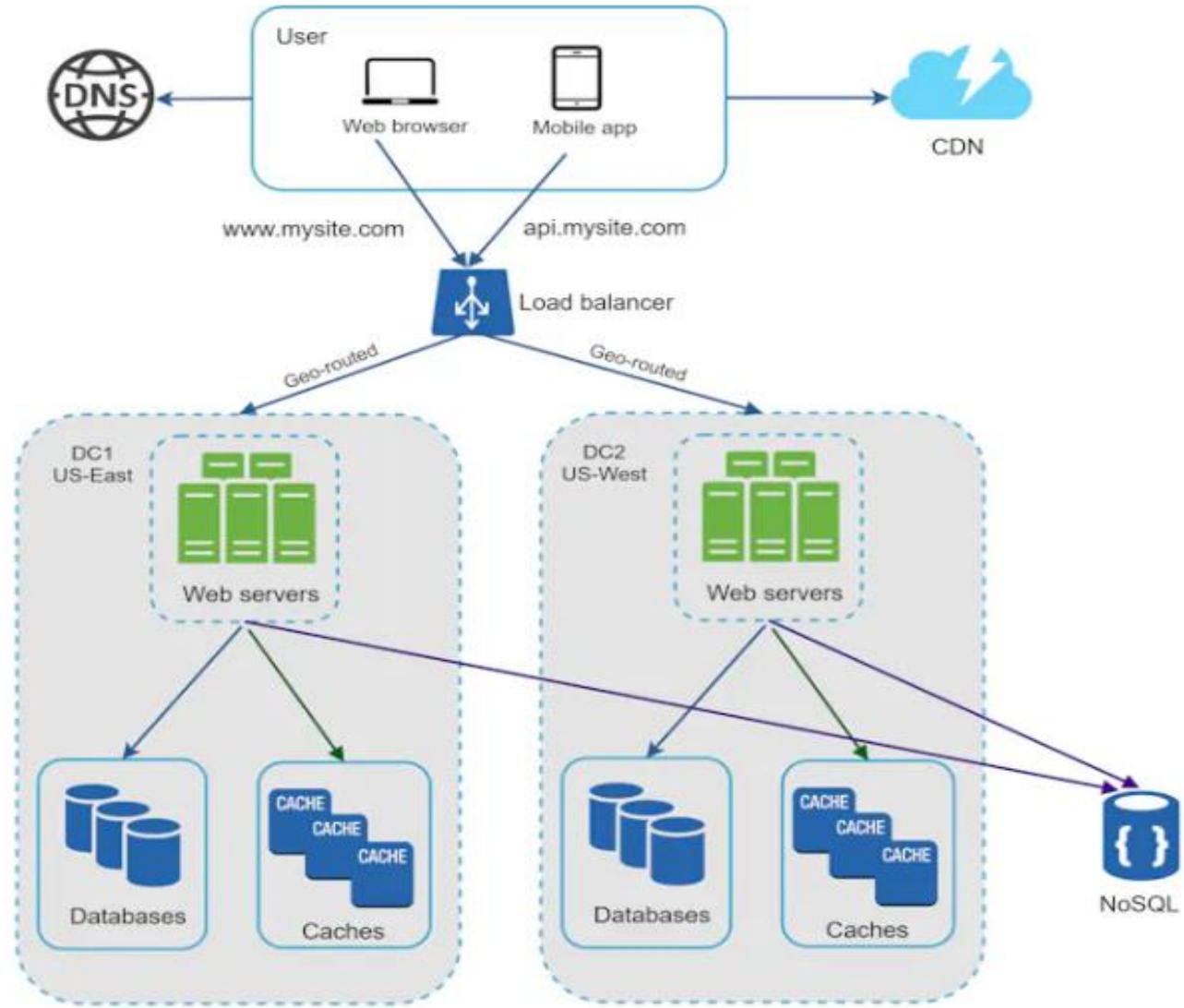


Taken from <https://bytebytego.com/courses/system-design-interview/scale-from-zero-to-millions-of-users>

# SCALE FROM ZERO TO MILLIONS

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## Data Centers



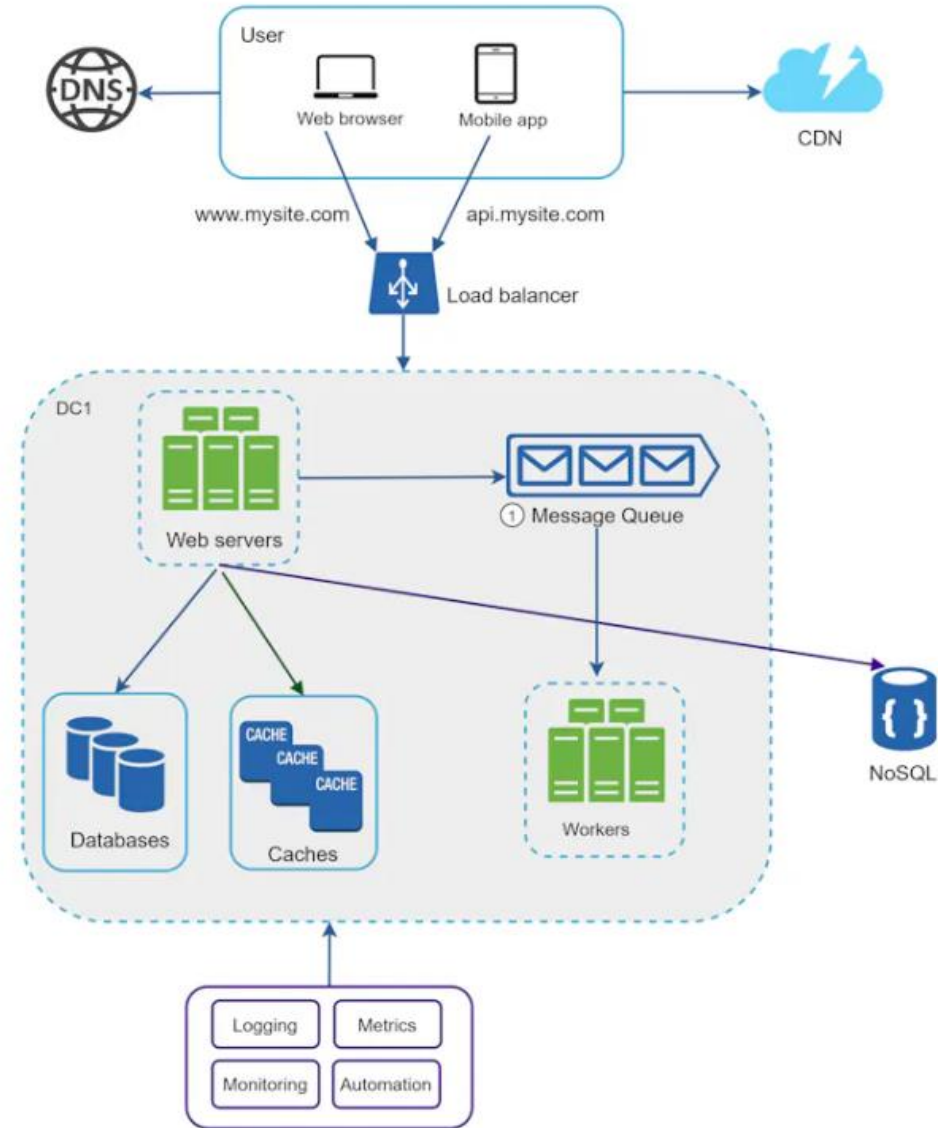
Taken from <https://bytebytego.com/courses/system-design-interview/scale-from-zero-to-millions-of-users>

# SCALE FROM ZERO TO MILLIONS

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Message Queue

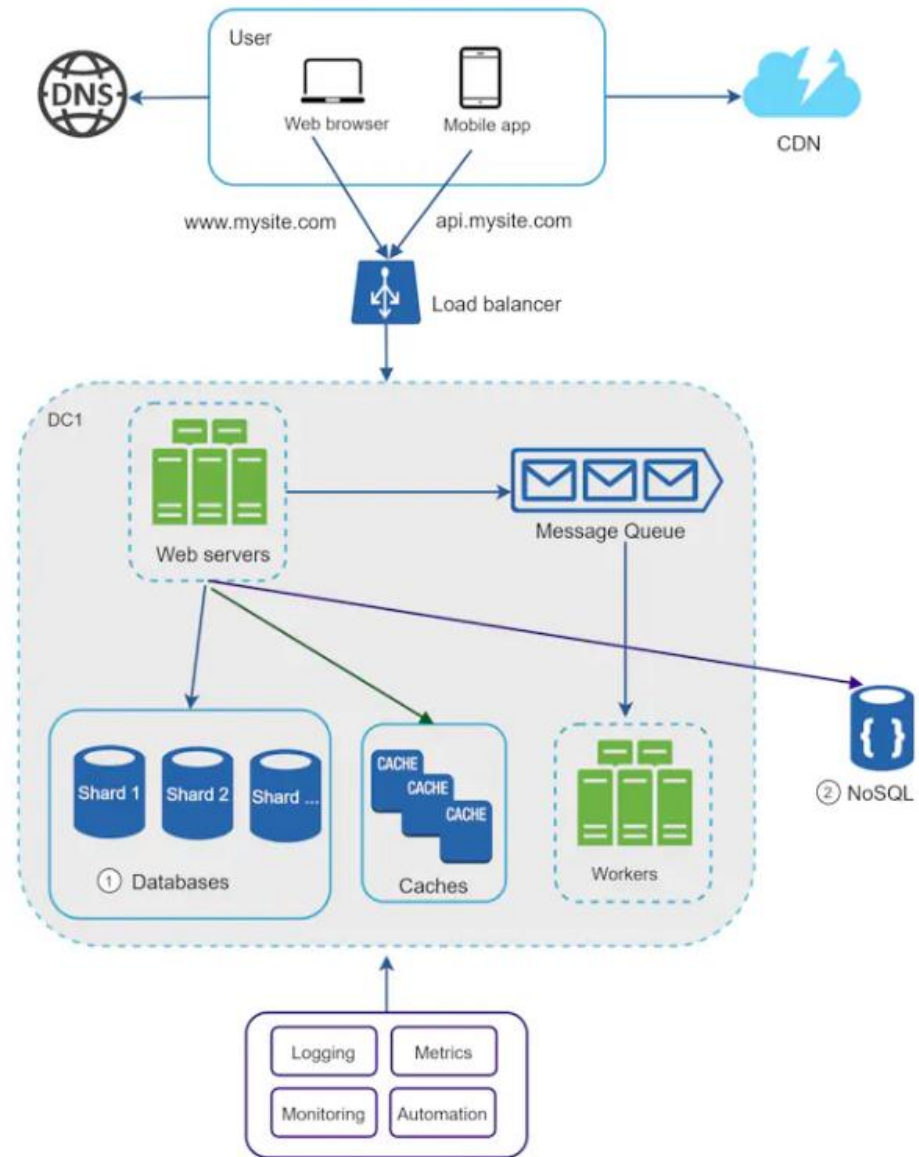
Logging and monitoring  
(and observability)



Taken from <https://bytebytego.com/courses/system-design-interview/scale-from-zero-to-millions-of-users>

# SCALE FROM ZERO TO MILLIONS

## Sharding



Taken from <https://bytebytego.com/courses/system-design-interview/scale-from-zero-to-millions-of-users>